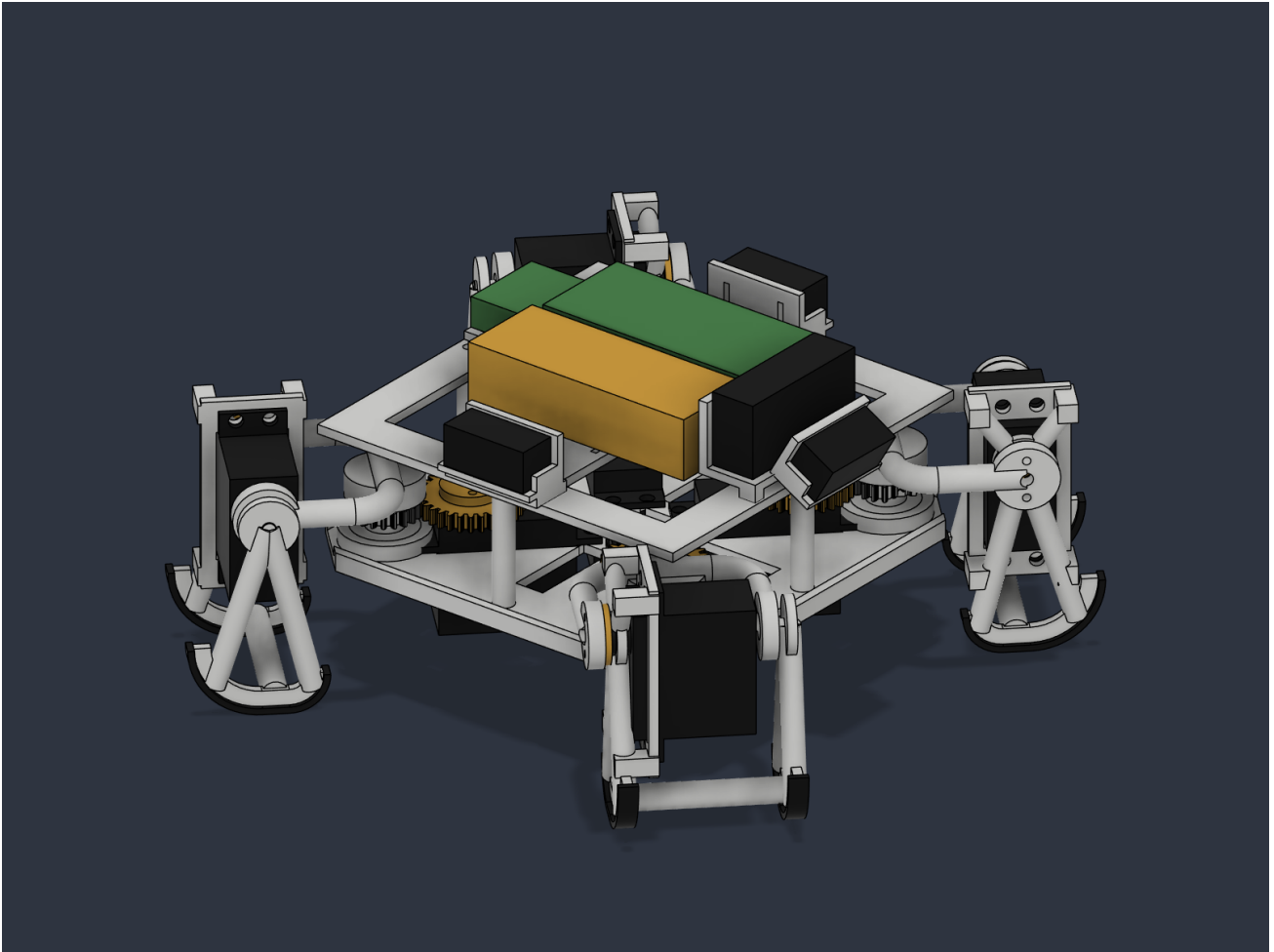


Build the B4 robot

Assembly order, component identification and unresolved interfaces



Use the reinforced B4 geometry

B4 retains the B3 rocker layout but adds four corner connections to each pitch-servo mount. The mount plate is nominally 3 mm thick; bridges are 6 mm wide and upper diagonal ribs are 6 mm in diameter.

Assembly sequence

1. Inspect and identify parts.
2. Fit one pitch-servo/rocker assembly.
3. Add its Y-fork and passive axle.
4. Fit yaw drive to the chassis.
5. Repeat for four legs.
6. Fit deck, sensors and wiring.
7. Check motion, then perform supported power-on tests.

What this guide can and cannot specify

This is an assembly guide for the existing prototype CAD, not a released production drawing. Servohorn attachment, bearing retention, several fastener lengths and the power system are not finalized. The relevant steps identify these gaps instead of assuming a working connection. Build one leg first.

Part list: structure

IDs are used throughout this guide. Quantities are for one complete robot.

ID	Part	Qty	Identification / fit
S01	Integrated chassis	1	Four yaw positions and four deck posts
S02	Electronics deck	1	Open plate with strap slots
S03	30T yaw driver	4	One per yaw servo; horn adapter unresolved
S04	Y-fork with integral 20T gear	4	Two arms support the pitch assembly
S05	Reinforced rocker / servo harness	4	B4 only: four corner ties and thicker plate
S06	IMU mounting pad	1	Flat pad with cable-tie slots
S07	Camera cradle	1	Open tray and upright back
S08	Left IR cradle	1	Side-facing holder
S09	Right IR cradle	1	Opposite side-facing holder
S10	Forward/down IR cradle	1	Angled holder at front
T01	Rubber tread strips	8	Two per rocker; nominal 2 mm thickness

Count before starting

19 structural parts + 8 tread strips. The four leg assemblies share the same arrangement. Keep the FL, RL, FR and RR labels from the CAD inventory so each part can be traced back to its source.

B4 identification

The reinforced rocker has four visible connections between its rectangular motor mount and the leg frame. Do not mix the older B3 rocker into a B4 set. The Y-fork clearance relief also changed slightly; use the B4 fork with the B4 rocker.

Part list: purchased items

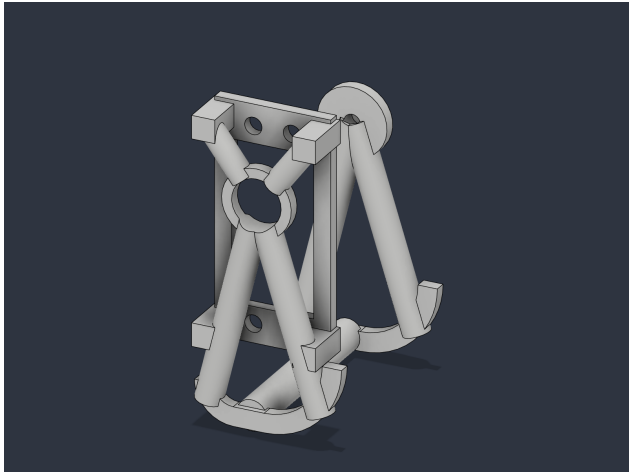
CAD envelopes represent space reservations; they are not detailed supplier drawings.

ID	Item	Qty	Status
M01	Parallax 900-00005 yaw servo	4	Provided servo type
M02	Parallax 900-00005 pitch servo	4	Same type; moves with rocker
H01	Yaw bearing, 12 x 21 x 5 mm	4	Nominal CAD envelope; verify actual bearing
H02	Yaw retention washer / screw set	4 sets	Washer modeled; screw and locking method unresolved
H03	Passive pitch bushing, 4 x 8 x 4 mm	4	Nominal ID x OD x length
H04	Passive M4 pitch axle	4	CAD placeholder; length and axial lock unresolved
H05	Pitch servo horn	4	Stock-horn envelope only; verify spline and holes
H06	Yaw servo horn / drive adapter	4	Required interface; not a separate modeled solid
E01	Sharp GP2Y0A41SK0F IR sensor	3	Left, right and forward/down
E02	MIKROE-4228 / MC6470 board	1	Accelerometer + magnetometer; no gyro
E03	M5Stack TimerCamera-X	1	Forward camera
E04	Battery	1	80 x 30 x 20 mm reservation; pack not selected
E05	Controller / driver / regulator	1 layout	80 x 35 x 18 mm reservation, not one specified board

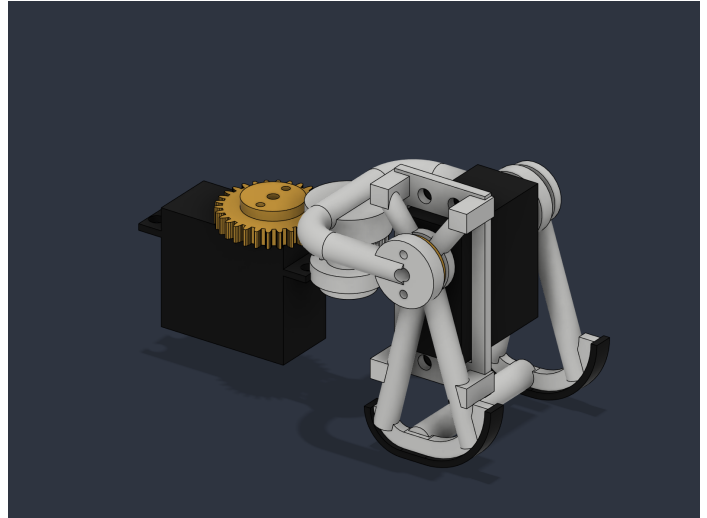
Also required: servo mounting fasteners, horn screws, deck fasteners, suitable retention hardware, straps/ties, tread adhesive, wiring, connectors and power protection. These are not a complete CAD bill of materials; confirm sizes and quantities during the first assembly. Do not order screw lengths from the rendered pictures.

Build one pitch assembly

The motor casing attaches to the rocker; its horn attaches to the Y-fork.



S05: reinforced rocker and four corner ties



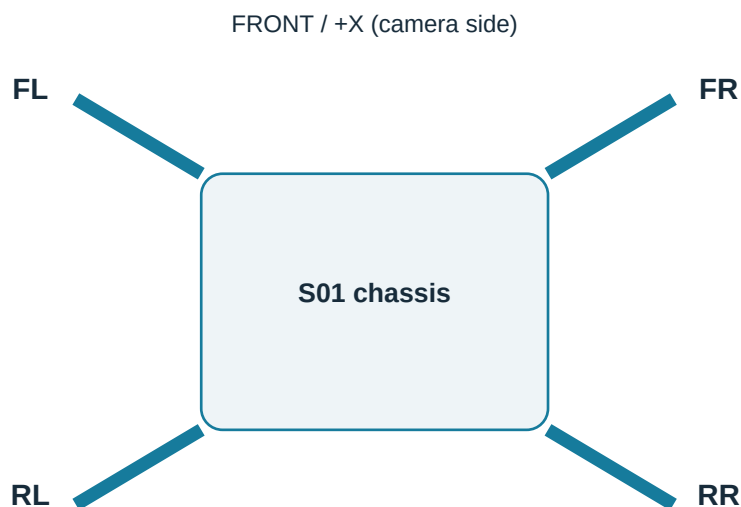
S04 + M02 + S05: assembled leg reference

- 1 Fit the pitch servo M02 into S05**
 Check the casing, mounting ears and cable exit against the opening. Keep the motor aligned along the leg. Use the four mounting holes, with suitable fasteners; do not distort the 3 mm plate by overtightening.
- 2 Center the servo, then attach horn H05**
 Establish the servo neutral position with your controller before installing the horn. The horn must fit the actual spline and its retaining screw. In this design the horn is fixed to the servo-side Y-fork arm, while the servo casing moves with the rocker.
- 3 Install the opposite support**
 Place H03 in the opposite fork eye and align it with the rocker pivot. Fit the passive H04 axle without clamping the moving joint solid. Its axial locking arrangement must be finalized before operation.
- 4 Check the complete pitch joint by hand**
 Support the leg and move gently without forcing the servo geartrain. Confirm casing, ribs, fork and wiring clear. Add two T01 tread strips per rocker, keeping their edges clear of moving parts. Repeat only after this first assembly fits.

Install yaw drives and legs

Repeat for FL, RL, FR and RR. Build the yaw system before fitting the upper deck.

- 5 Mount yaw servo M01 to chassis S01**
 The output shaft points upward; the casing projects below the chassis. Use all four mounting-ear holes. Keep wires away from the gears and from the leg swing.
- 6 Fit bearing H01 and the Y-fork S04**
 The vertical fork shaft locates in the bearing at the corner. CAD uses a nominal 12 mm shaft and 21.1 mm bearing pocket. Verify fit and a positive axial retention method: a washer alone must not be assumed to secure both bearing rings.
- 7 Fit the retention set H02**
 The CAD shows a washer below the yaw shaft and a screw allowance. Determine the actual screw engagement, bearing seat and axial clearance on the first unit. The fork must rotate freely with no lift-out or significant play.
- 8 Connect driver S03 to its servo horn H06**
 The 30T gear engages the fork's 20T gear. The horn-to-gear attachment is not finalized in CAD; match the real horn and provide a secure connection. Check mesh throughout the intended yaw sweep without binding.
- 9 Align neutral leg directions**
 With each yaw servo at neutral, point its leg diagonally outward. Do not command the servo beyond its actual travel to achieve the larger yaw range once requested for this robot.



Top-view orientation only; not to scale. Left is +Y in CAD.

Deck, sensors and wiring

Install after all four mechanical leg assemblies fit and move freely.

10 Fit deck S02 to the four chassis posts

Check cable access before securing the deck. Its nominal holes are 3.4 mm and the post pilot holes are 2.6 mm. Finalize the thread/insert and screw length for the actual joint; these dimensions do not specify a validated fastener.

11 Fit the camera and IR holders

S07 supports forward-facing E03. S08/S09 hold E01 sensors facing left and right. S10 holds the third E01 facing forward/down. The current design uses open cradles with strap slots, not verified sensor mounting-hole patterns.

12 Fit IMU E02 to pad S06

Secure it rigidly and record its axis orientation for software. Keep it clear of moving joints. The MC6470 includes an accelerometer and magnetometer, not a gyroscope. Choose a location and wiring that can be checked for magnetic interference.

13 Secure battery E04 and electronics E05

The yellow and green boxes in CAD are reserved volumes. Select actual hardware before final mounting. Use the deck strap slots where suitable; provide insulation and a secure attachment for every board. The current B4 battery reservation sits above the deck.

14 Route and label every cable

Label servo cables FL-yaw, FL-pitch, FR-yaw, FR-pitch, RL-yaw, RL-pitch, RR-yaw and RR-pitch. Leave a flexible loop for pitch and yaw, secure the fixed ends, and check that no cable can enter a gear or be pulled taut.

Electrical integration is still a separate design step

Use a servo supply matched to the actual servo specification and a regulator matched to the controller. Do not power eight servos through a controller GPIO or small board regulator. Establish the required common signal ground for non-isolated control, verify connector polarity, and size wiring/protection for the selected system.

This guide does not assign connector pins, fuse ratings, battery voltage or regulator capacity: the project has not selected a complete power system. Verify those details before connecting the battery.

Final checks and open items

Complete one supported leg test before attempting weight-bearing locomotion.

Check	Acceptance before operation
Mechanical retention	Servo ears secure; horns locked to the correct splines; yaw shaft, bearing and passive axle cannot work loose.
Freedom of movement	No binding or rubbing during a slow sweep; no screw or washer clamps a rotating interface.
Cable movement	Cable loops remain slack enough throughout the intended range and stay outside gear teeth.
Initial power-on	Support the chassis with legs clear of the surface. Test one servo at a time, starting near neutral with small movements.
Direction and limits	Record each joint's neutral, direction and usable range. Stop at rubbing, abnormal noise or excessive current; do not force a hard stop.
Ground test	Check all four feet contact evenly, tread attachment holds, and the mount ties show no visible damage or excessive deflection.

Resolve before calling the assembly complete

- 1. Real horn dimensions and positive 30T driver attachment.
- 2. Yaw-bearing seating and axial retention.
- 3. Passive M4 axle length and locking arrangement.
- 4. Screw lengths, thread/insert choices and holder attachment.
- 5. Actual battery, controller, regulators and wiring diagram.

Validation scope

B4 has four verified solid bridge cores and remains connected geometry. Sampled pitch checks from -45 to +45 degrees passed; neutral-pitch yaw samples within +/-45 degrees passed. Larger yaw angles still collide with fixed components. Continuous combined motion, adjacent-leg motion, load capacity and cable clearance are not certified. Existing learned simulations still use B3.

Source files and traceability

Based on **cad/B4_reinforced_harness/**:
Robot_B4_Reinforced_Harness.f3d / .step; B4_Reinforced_Leg_Detail.f3d; B4_Reinforced_Rocker.f3d; B3_inventory.json; harness_ties.json; B3_clearance_audit.json.
The B3-prefixed inventory names are inherited component labels inside B4. Illustrations are renders of the B4 CAD. Quantities were checked against all 62 CAD bodies; required yaw horns and additional fasteners are listed separately because they are not fully modeled.